Internship-Portal

A PROJECT REPORT

Submitted by

Shreyansh Jain (2303051050882)
Lalee Kumawat (2303051050393)
Anand Solanki (2303051050911)
Parmar Himanshu sinh (2303051050507)

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DR KHUSHBOO TRIVEDI

Associate Professor

Computer Science &Engineering Department

Parul Institute of Technology



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PARUL UNIVERSITY

PARUL INSTITUTE OF TECHNOLOGY



CERTIFICATE

This is to certify that Project-II (**303105423**) of 7th Semester entitled "Internship-Portal" of Group No. 7 has been successfully completed by

Shreyansh Jain – 2303051050882

Lalee Kumawat – 2303051050393

Anand Solanki – 2303051050911

Parmar Himanshu Sinh – 2303051050882

under my guidance in fulfillment of the Bachelor of Technology (B.TECH.) in Computer Science & Engineering of Parul University in Academic Year 2026-2027

Dr. Khushboo Trivedi — **Project Guide**

DR. ARNIKA PATEL — **Project Coordinator**

Dr. SUMITRA MENARIA — **Head of Department**

CSE

EXTERNAL EXAMINER

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Behind any major work undertaken by an individual there lies the contribution of the people who helped him to cross all the hurdles to achieve his goal.

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Last but not the least, our humble thanks to the Almighty God.

Place: Vadodara

ABSTRACT

Internship-Portal is an enterprise-grade internship discovery, resume engineering, and hiring management platform designed to reduce friction in modern internship recruitment. Built on the MERN technology stack, the platform connects students or candidates, verified employers or recruiters, and system administrators through a unified web application.

The platform provides dual-mode authentication using stateless JWT authentication and Google OAuth 2.0, role-based access control, internship discovery and filtering, one-click applications, application lifecycle tracking, an interactive AI resume builder, conversational AI assistance, employer verification, and transactional email notifications through Nodemailer.

The system follows a decoupled three-tier architecture using React 18 and Vite at the presentation tier, Node.js and Express.js at the application tier, and MongoDB 7.0 at the data tier. The application is containerized with Docker and Docker Compose and deployed on AWS EC2 behind an Nginx reverse proxy. Security hardening includes Helmet, CORS controls, rate limiting, password hashing, and protected role-based routes.

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LIST OF ABBREVIATIONS

S.No.	Abbreviation	Full Form
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	AWS	Amazon Web Services
4	CORS	Cross-Origin Resource Sharing
5	CSP	Content Security Policy
6	DFD	Data Flow Diagram
7	EC2	Elastic Compute Cloud
8	FCP	First Contentful Paint
9	GUI	Graphical User Interface
10	HMR	Hot Module Replacement
11	HTTP	Hypertext Transfer Protocol
12	JWT	JSON Web Token
13	MERN	MongoDB, Express, React, Node.js
14	Nginx	Engine X (web server / reverse proxy)
15	NoSQL	Not Only SQL
16	OAuth	Open Authorization
17	RBAC	Role-Based Access Control
18	REST	Representational State Transfer
19	SMTP	Simple Mail Transfer Protocol
20	SPA	Single-Page Application

CHAPTER 1
INTRODUCTION

1.1 Internship-Portal — Overview

Internship-Portal is a unified internship discovery and hiring management platform developed to address fragmentation in the internship recruitment process. It brings candidate discovery, interactive resume creation, employer management, application tracking, and administrative governance into a single cloud-based application.

The platform supports three authenticated personas: Students/Candidates, Verified Employers/Recruiters, and System Administrators. Students can discover internships, maintain profiles, build resumes, apply to opportunities, and track application progress. Employers can create listings, review applicants, and update application states. Administrators verify employers and monitor ecosystem-level information.

1.2 Problem Definition

The contemporary internship and recruitment landscape is affected by fragmented platforms, untracked application lifecycles, inefficient recruitment pipelines, and limited real-time transparency between candidates and hiring teams.

Traditional hiring boards often depend on static forms and disconnected communication channels. This creates candidate churn and increases administrative overhead for employers. Applicants may have difficulty tracking application progress, while employers need to manage listings and candidate states across separate workflows.

1.3 Motivation

Internship-Portal was motivated by the need to unify internship discovery and hiring operations in one system. The project focuses on reducing friction for candidates while giving employers and administrators structured tools for recruitment management.

The platform combines internship search, candidate profiles, resume generation, employer verification, application state management, AI career assistance, and transactional notifications.

This integration provides a single workflow rather than a collection of disconnected recruitment activities.

1.4 Objectives

The primary technical and operational objectives realized in Internship-Portal are:

- To provide a single-portal ecosystem supporting students/candidates, verified employers/recruiters, and system administrators.
- To build a responsive frontend using React 18 and Vite with mobile responsiveness and a consistent design system.
- To provide a decoupled RESTful API using Express.js for authentication, role management, internship listings, applications, and status transitions.
- To implement dual-mode authentication using bcrypt-based password authentication and Google Identity Services OAuth 2.0.
- To implement an interactive AI Resume Builder with real-time preview and PDF/Print export.
- To provide employer verification and administrative governance for maintaining platform integrity.

To containerize and deploy the application using Docker, Docker Compose, Nginx, and AWS EC2.

1.5 Scope of the Project

1.5.1 Existing System

The existing recruitment approach described in the Internship-Portal report is fragmented. Internship discovery, candidate information, application submission, communication, and recruitment management may depend on static forms and decoupled channels. Such an approach makes it difficult to maintain a unified view of the application lifecycle.

1.5.2 Proposed System

Internship-Portal proposes a unified web platform where students can search and apply for internships, employers can publish and manage opportunities, and administrators

can verify employers and monitor the platform. The system combines authentication, listing management, application tracking, resume generation, AI assistance, notifications, and cloud deployment in one architecture.

1.6 Hardware and Software Requirements

Table 1.1 – Hardware and Software Requirements

Category	Requirement	Specification
Hardware	Cloud Host	AWS EC2; instance size not specified in source
Hardware	Operating System	Ubuntu 24.04 LTS
Software	Frontend	React.js 18.2, Vite 5.x, CSS3
Software	Backend	Node.js 20 LTS, Express.js 4.19
Software	Database	MongoDB 7.0, Mongoose 8.x
Software	Authentication	JWT, bcryptjs, Google Identity Services OAuth 2.0
Software	Security	Helmet, CORS, express-rate-limit
Software	Reverse Proxy	Nginx Alpine
Software	Containerization	Docker & Compose v2
Software	Email	Nodemailer / SMTP
Software	Cloud	AWS EC2

The provided engineering report specifies the production infrastructure and software stack, but does not state a local minimum RAM, CPU, or disk requirement. The deployment environment is AWS EC2 running Ubuntu 24.04 LTS, with Docker-based services and MongoDB persistence.

CHAPTER 2
LITERATURE SURVEY

2.1 Review of Existing Recruitment Approach

The Internship-Portal report identifies fragmentation as a central weakness in contemporary internship recruitment. Traditional hiring boards rely on static forms and separate communication channels, which can result in candidate churn and administrative overhead.

The project addresses this gap by consolidating discovery, profile management, application processing, resume engineering, employer verification, and application pipeline tracking into a single system. The report does not provide a formal academic literature review or named competing platforms, so no additional external comparison is introduced here.

2.2 Advantages, Disadvantages and Limitations of Existing Approach

Table 2.1 – Existing Approach Analysis

System / Approach	What It Provides	Disadvantages	Limitations
Traditional hiring boards	Internship discovery and application collection	Static forms and decoupled communication	Application lifecycle can be difficult to track centrally
Fragmented recruitment workflow	Separate channels for candidates and hiring teams	Higher administrative overhead	No single unified recruitment view
Internship-Portal	Unified discovery, applications, resume builder, employer verification and administration	More components to operate than a simple listing form	Source lists HTTPS automation, ML matching, video interviews and messaging as future work

The existing approach offers basic internship discovery and application collection, but the report identifies limitations around fragmented workflows, application lifecycle visibility, and administrative overhead. These limitations motivate the unified architecture of Internship-Portal

.

2.3 Background Technologies and Proposed System Analysis

2.3.1 React.js and Vite

React.js 18.2 is used for the component-based presentation layer, with declarative state, hooks, and context API. Vite 5.x provides hot module replacement during development and optimized Rollup-based production bundling.

2.3.2 Node.js and Express.js

Node.js 20 LTS provides the event-driven asynchronous backend runtime. Express.js 4.19 supplies RESTful routing, middleware processing, error management, authentication protection, and application-level business logic.

2.3.3 MongoDB and Mongoose

MongoDB 7.0 is used as the document-oriented NoSQL data store. Mongoose 8.x provides schema definitions, pre-save hooks, referential integrity, and connection management for the application models**.**

2.3.4 Docker, Docker Compose and Nginx

Docker and Compose v2 provide isolated and deterministic service environments. Nginx Alpine acts as the production reverse proxy and static edge server, handling public traffic, static assets, compression, caching, and API proxying.

2.3.5 Authentication and Security Technologies

The platform uses bcryptjs for password hashing, Google Identity Services for federated authentication, JWT for stateless sessions, Helmet for security headers, CORS controls for cross-origin protection, and express-rate-limit to restrict authentication attempts.

2.3.6 AWS EC2 Cloud Infrastructure

The production application is provisioned on Amazon Web Services Elastic Compute Cloud using Ubuntu 24.04 LTS. The report specifies public HTTP access through port 80 and SSH management through port 22.

2.3.7 Advantages of the Proposed System

Unified internship discovery and hiring workflows in one portal.

Role-based experiences for students, employers, and administrators.

Dual authentication using JWT and Google OAuth 2.0.

Application lifecycle tracking with explicit application states.

Interactive resume generation with real-time preview and export.

Containerized deployment with isolated internal services.

Nginx reverse proxy and caching for production delivery.

Employer verification before internship listings are published.

2.3.8 Limitations of the Proposed System

The provided report does not specify a formal load-testing methodology or detailed benchmark dataset.

- HTTPS/SSL automation is listed as a future enhancement, so the current report documents HTTP public access on port 80.
- Semantic machine-learning matching between resumes and internship requirements is listed as future work rather than a current capability.
- Video interview scheduling and real-time direct messaging are not yet part of the implemented feature set described in the report.

CHAPTER 3
METHODOLOGY

3.1 Existing Methodology

The existing internship recruitment workflow can be represented as a sequence of discovery, profile submission, application, employer review, and communication activities performed across fragmented channels. The Internship-Portal report specifically highlights static forms, decoupled communication, untracked application lifecycles, and administrative overhead as weaknesses of this approach.

Because the provided source report does not define a formal legacy workflow diagram, the comparison below is limited to the problems explicitly described in the source.

3.2 Proposed Methodology

Internship-Portal uses a decoupled three-tier client-server architecture. Public requests reach Nginx, which serves the React single-page application and proxies API calls to the internal Node.js/Express service. The application service authenticates users, enforces roles, processes internship and application operations, and connects to MongoDB through Mongoose.

3.2.1 Phase 1 — Authentication and Role Management

Table 3.2 – Authentication and Security Controls

Authentication / Security	Implementation	Purpose	Result	Access
Password Authentication	bcryptjs	Password hashing	Verified credentials	Public login
JWT Authentication	Signed JWT	Stateless session	Authenticated token	Private routes
Google OAuth 2.0	Google Identity Services	Federated identity	Verified Google credential	Public login
RBAC	Express middleware	Role restriction	Admin/company/student authorization	Protected routes

A user registers or signs in using password authentication or Google OAuth 2.0. The backend validates credentials and issues a cryptographically signed JWT. Protected routes use middleware

to validate the Bearer token, while authorization middleware restricts controller access based on roles such as admin or company.

3.2.2 Phase 2 — Internship Discovery

Students access active internship listings through the public internship API. Opportunities can be searched and filtered using keywords, location, internship type, stipend bands, category, and required skills. The student can open a listing and review the available role information before applying.

3.2.3 Phase 3 — Application Processing

A student submits an application with the relevant resume and cover letter. The Application model links the internship to the student and records the lifecycle state. The supported application states are applied, shortlisted, accepted, and rejected.

3.2.4 Phase 4 — Employer Recruitment Pipeline

Employers create, preview, edit, and deactivate internship listings. They can view submitted applications, review candidate qualifications and resumes, and transition candidate states. Database synchronization makes these changes available to the application workflow.

3.2.5 Phase 5 — Administrative Verification

Administrators review pending company registrations and approve or suspend employer accounts. The verification gate prevents unapproved companies from publishing internship listings according to the system design described in the report.

3.2.6 Phase 6 — Resume Engineering and AI Assistance

The interactive resume builder allows candidates to create ATS-optimized resumes with real-time preview, typography selection, skill formatting, and PDF/Print export. The platform also exposes an AI chat endpoint for career assistance and navigation queries.

3.2.7 Phase 7 — Notification and Deployment Flow

Table 3.3 – User Roles and Capabilities

Role	Key Capabilities
Student	Browse/filter internships, manage profile, build resume, apply, track applications
Employer	Create/manage listings, review applicants, inspect resumes, update application status
Admin	Verify/suspend companies, view ecosystem analytics, monitor system health

Dimension	Existing Approach	Proposed Methodology (Internship-Portal)
Discovery	Fragmented / static channels	Unified internship discovery and filtering
Application Tracking	Lifecycle may be untracked across channels	Explicit applied / shortlisted / accepted / rejected states
Employer Verification	Separate administrative process	Integrated admin approval gate
Resume Preparation	Separate candidate activity	Interactive resume builder with preview and export
Authentication	Disconnected login workflows	JWT plus Google OAuth 2.0
Deployment	Not unified in source description	Docker Compose + Nginx + AWS EC2

Nodemailer provides transactional alerts for application and platform updates. The complete ecosystem is containerized using Docker Compose, with the React client served by Nginx, the Node.js backend on internal port 5000, and MongoDB on internal port 27017.

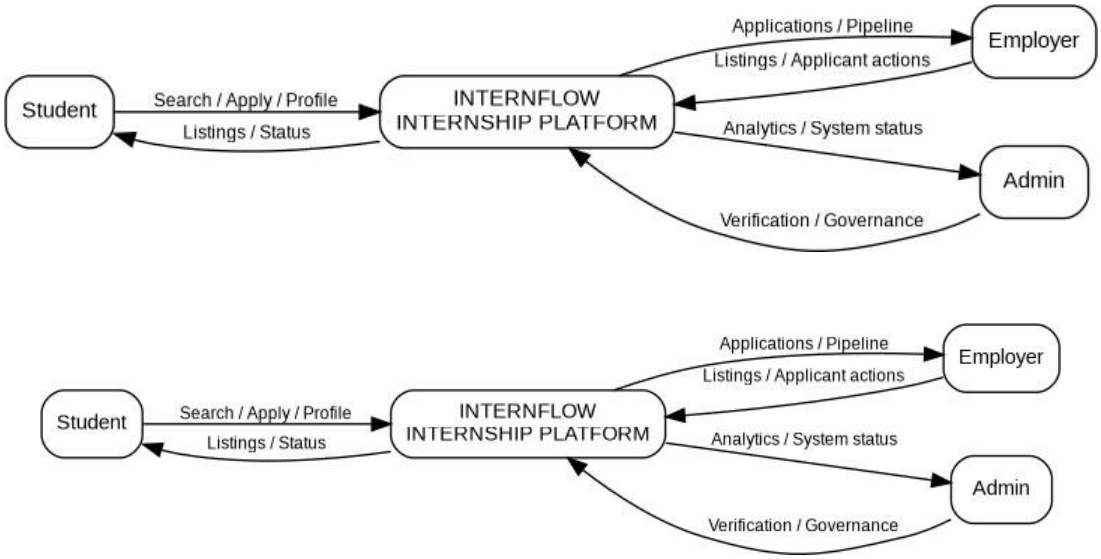
CHAPTER 4
SYSTEM DESIGN AND DIAGRAMS

4.1 Data Flow Diagram (DFD)

4.1.1 Level 0 DFD — Context Diagram

The Level 0 DFD represents Internship-Portal as a single platform interacting with three principal external actors: Student, Employer, and Admin. Students request internship information and submit applications. Employers manage listings and candidate states. Administrators perform employer verification and monitor platform information.

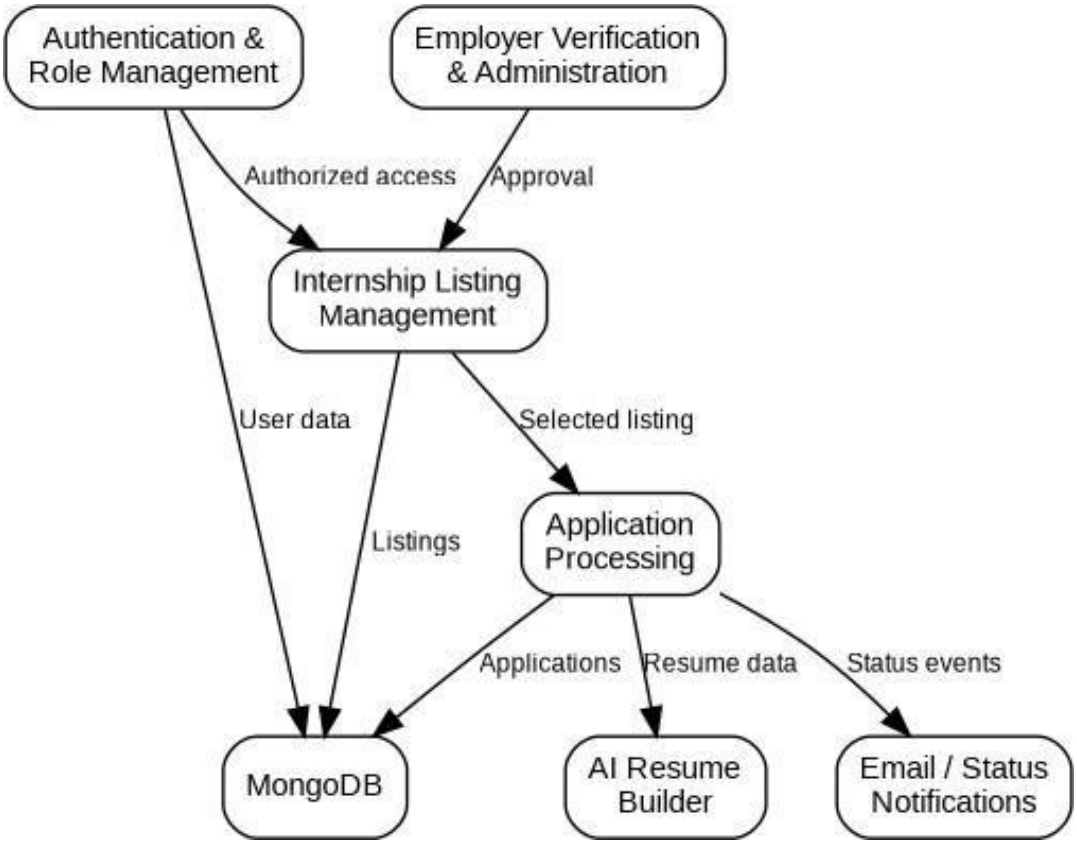
Figure 4.1 – Level 0 DFD — Context Diagram

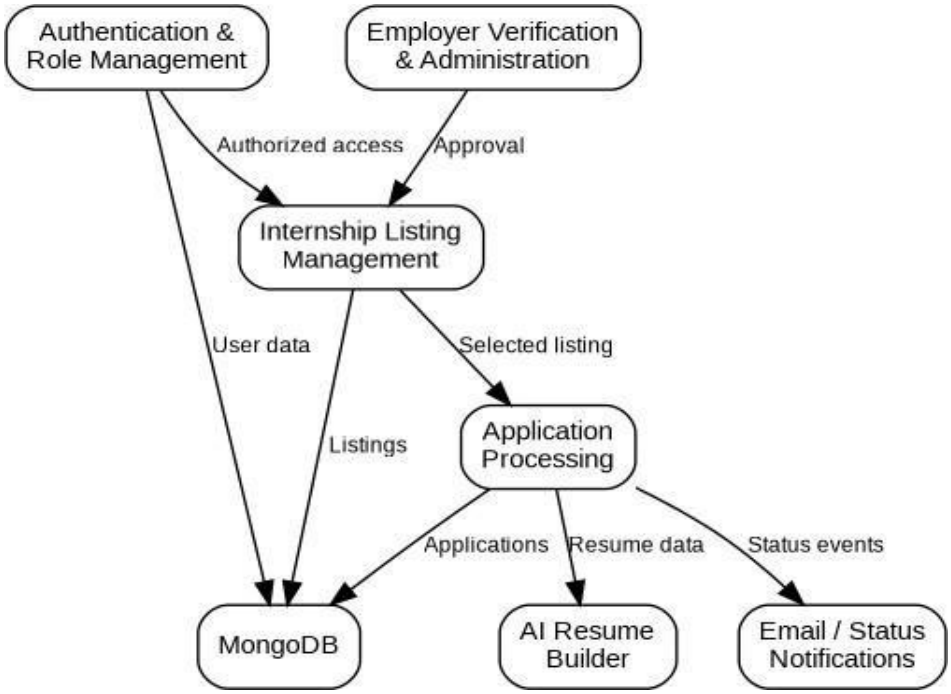


4.1.2 Level 1 DFD — Platform Workflow

The Level 1 DFD decomposes the platform into authentication and role management, internship listing management, application processing, resume generation, administrative verification, notification services, and MongoDB persistence.

Figure 4.2 – Level 1 DFD — Platform Workflow

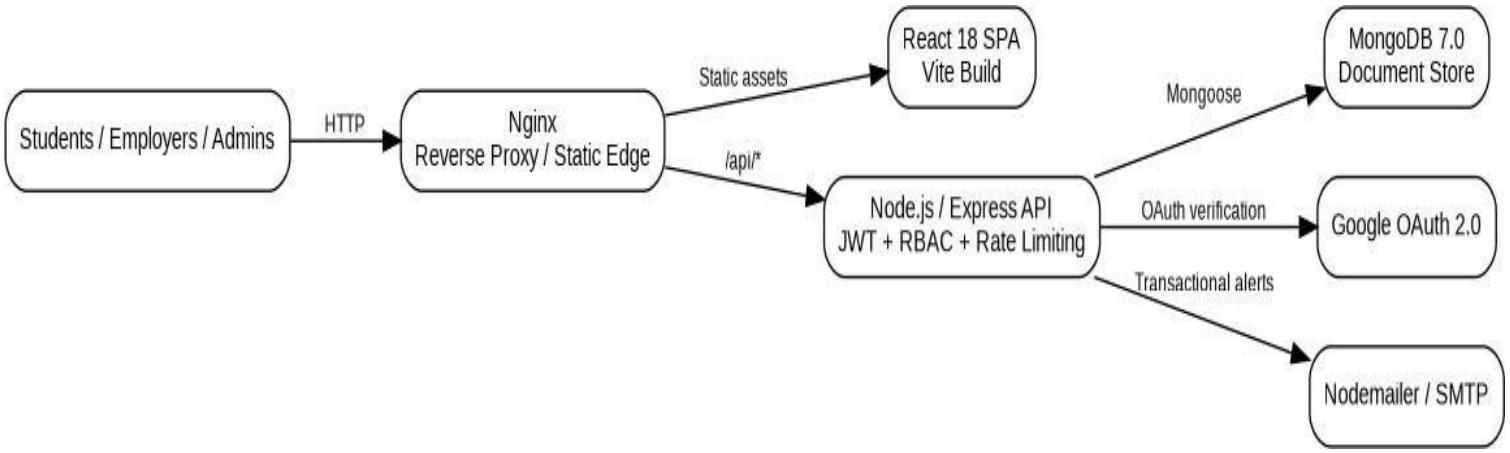




4.1.3 System Architecture

The production architecture uses Nginx as the public edge, React 18 as the presentation tier, Express.js and Node.js as the application tier, and MongoDB 7.0 as the data tier. Google OAuth 2.0 provides identity services and Nodemailer provides transactional alerts.

Figure 4.3 – Internship-Portal Production Multi-Tier Cloud Architecture

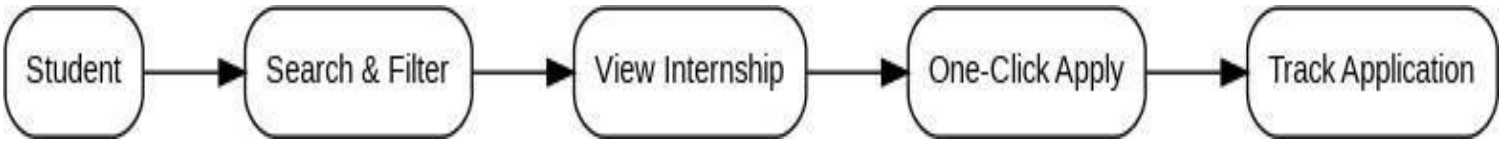


4.2 Use Case Diagram

4.2.1 Student Use Cases

The Student actor can register, log in, browse internships, search and filter listings, view internship details, apply, update profile information, build a resume, and track application status.

Figure 4.4 – Student Use Case / Application Flow



4.2.2 Employer Use Cases

The Employer actor can register, log in, create and manage internship listings, view submitted applications, inspect candidate profiles and resumes, and update application states such as shortlist, accept, or reject.

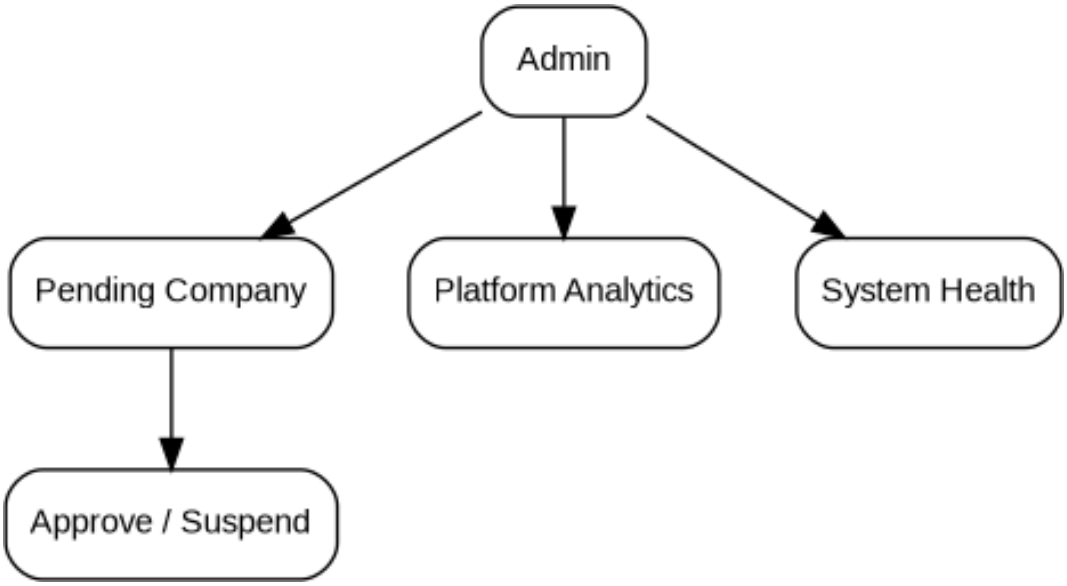
Figure 4.5 – Employer Recruitment Pipeline



4.2.3 Admin Use Cases

The Admin actor can review pending companies, approve or suspend employers, view ecosystem analytics, and monitor system health.

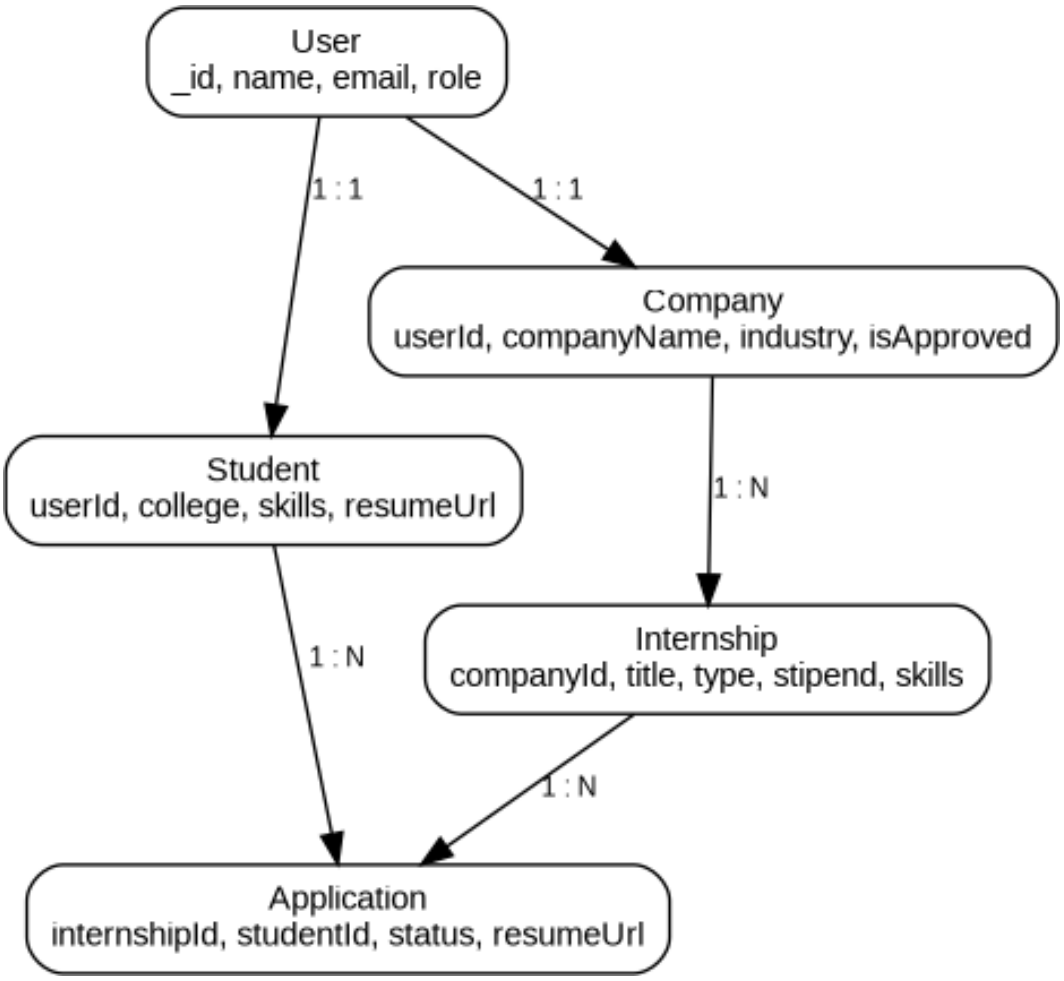
Figure 4.6 – Administrative Control Flow



4.3 Entity-Relationship Diagram

The Internship-Portal database is organized around User, Student, Company, Internship, and Application models. User acts as the root identity model. Student and Company profiles reference User. Approved companies create Internship listings, while Student records are connected to applications for those listings.

Figure 4.7 – Entity-Relationship Diagram



CHAPTER 5
IMPLEMENTATION

5.1 Module Description

Table 5.1 – Module Summary

Module	Technology	Description
Authentication Module	Node.js, Express, JWT, bcryptjs, Google OAuth	Registration, login, JWT sessions and federated authentication.
Internship Discovery Module	React, Node.js, MongoDB	Listing search, filtering and internship details.
Application Module	Node.js, Express, MongoDB	Application creation, materials and lifecycle state tracking.
Employer Module	React, Node.js, MongoDB	Listing management and applicant review.
AI Resume Builder	React.js	Interactive ATS-oriented resume creation, preview and export.
Administration Module	React, Node.js, MongoDB	Employer verification, analytics and system health.
Notification Module	Nodemailer / SMTP	Transactional alerts and application updates.
Deployment Module	Docker, Compose, Nginx, AWS EC2	Containerized production deployment and reverse proxying.

The implementation is organized into functional modules that correspond to the platform's main workflows. The technology stack and responsibilities are summarized in Table 5.1.

5.1.1 Authentication Module

The authentication module supports registration, password login, JWT issuance, Google OAuth 2.0 authentication, and authenticated session retrieval. Passwords are protected using bcryptjs hashing.

5.1.2 Internship Discovery Module

The discovery module exposes active internship listings and supports filtering by keywords, location, stipend bands, category, and required skills. Students can inspect listings before submitting applications.

5.1.3 Application Tracking Module

The application module records the internship, student, application materials, and application timestamp. Employer actions transition the application between applied, shortlisted, accepted, and rejected states.

5.1.4 Employer Management Module

Employers manage their company profile and internship listings. The administrative approval gate controls whether a company can publish listings.

5.1.5 AI Resume Builder Module

The resume builder provides real-time editing and preview, typography selection, skill formatting, and PDF/Print export. The source report describes it as an automated client-side interactive resume generator.

5.1.6 Administration Module

The administration center provides employer verification and ecosystem analytics, including active listings, total applications, candidate placement ratios, and system health information.

5.1.7 Notification Module

Nodemailer and SMTP services are used for transactional alerts and application updates.

5.1.8 Deployment Module

Docker multi-stage builds produce the production React client image. Docker Compose manages the client, server, and MongoDB services, while Nginx provides the public edge. The production environment runs on AWS EC2 with Ubuntu 24.04 LTS.

5.2 Pseudo Code of Modules

5.2.1 User Login and JWT Issuance

```
FUNCTION login(email, password):
  user = findUserByEmail(email)
  IF user does not exist:
    RETURN authentication error
  valid = comparePassword(password, user.password)
  IF valid is false:
    RETURN authentication error
  token = signJWT({ id: user.id, role: user.role }, expiry=30 days)
  RETURN token
```

5.2.2 Internship Application

```
FUNCTION applyForInternship(studentId, internshipId, resumeUrl, coverLetter):
  internship = findInternshipById(internshipId)
  IF internship is not active:
    RETURN application error
  application = createApplication(
    internshipId, studentId, resumeUrl, coverLetter,
    status="applied"
  )
```

```
  RETURN application
```

5.2.3 Employer Application Status Update

```
FUNCTION updateApplicationStatus(companyId, applicationId, newStatus):
  application = findApplicationById(applicationId)
  verifyEmployerOwnsInternship(companyId, application.internshipId)
  application.status = newStatus
  save(application)
  sendNotification(application.studentId, newStatus)
  RETURN application
```

5.2.4 Employer Verification

```
FUNCTION approveCompany(adminId, companyId):
  verifyRole(adminId, "admin")
  company = findCompanyById(companyId)
  company.isApproved = true
  save(company)
  RETURN approval confirmation
```

5.3 Platform Components and Deployment Views

The provided Internship-Portal report does not contain application screenshots. To keep this report structurally aligned with the the sample report template without presenting invented screenshots as real evidence, the figure section uses system-flow and deployment diagrams derived from the source architecture and functional descriptions.

Figure 5.1 – Interactive Resume Builder Flow

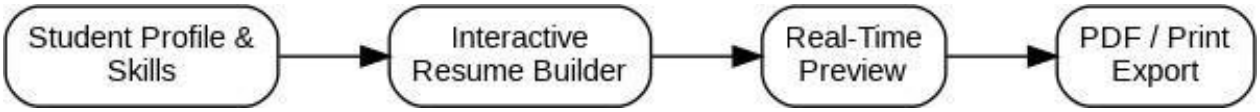


Figure 5.2 – REST API Gateway View

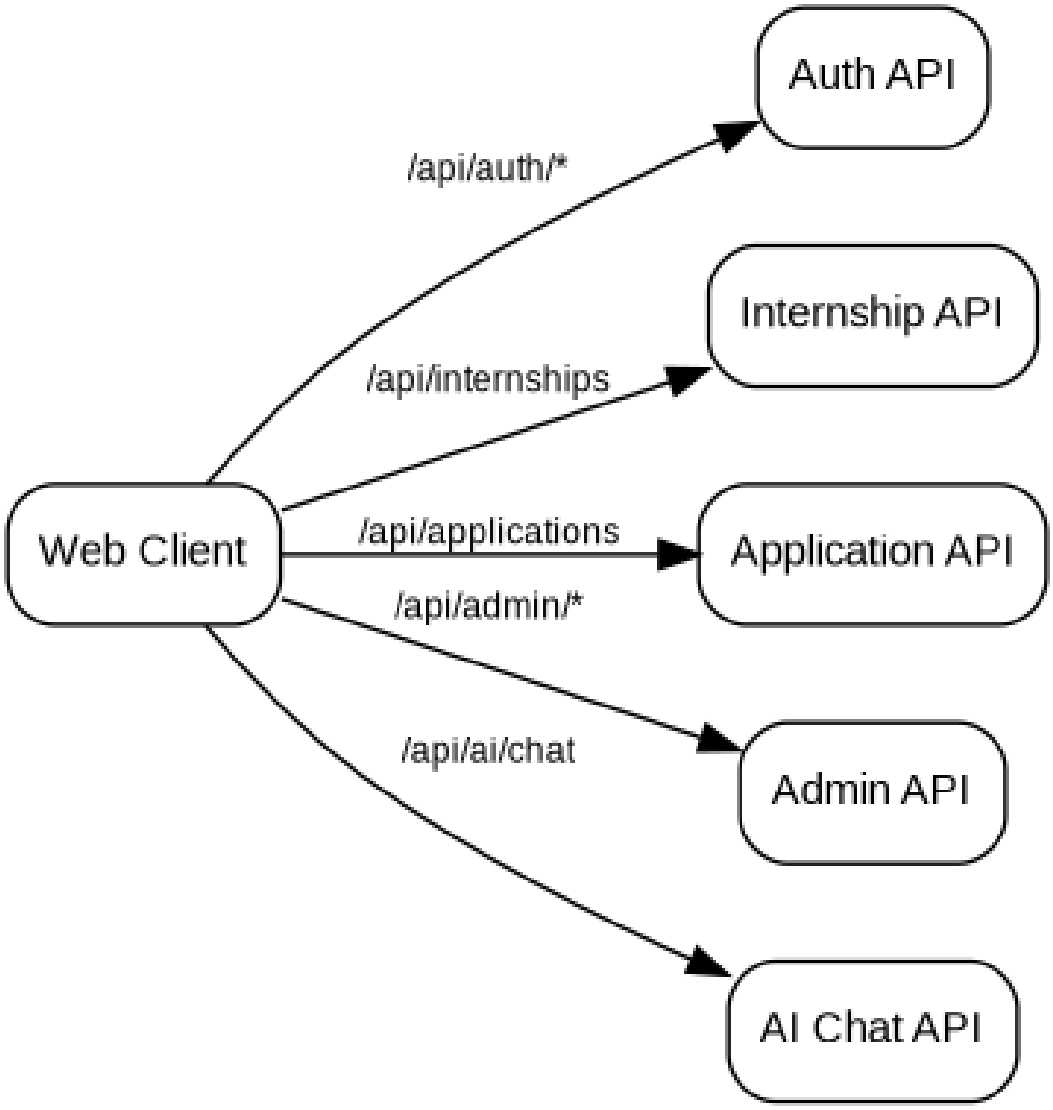


Figure 5.3 – Docker and AWS Deployment Flow

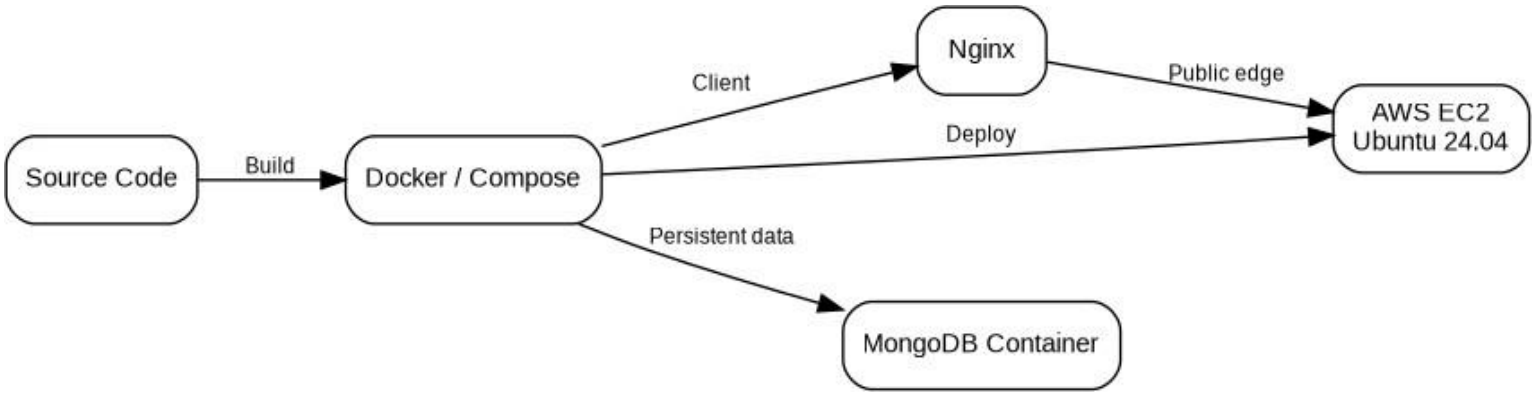


Figure 5.4 – Student Discovery and Application Flow



Figure 5.5 – Employer Recruitment Pipeline Flow



Figure 5.6 – Administrative Verification Flow

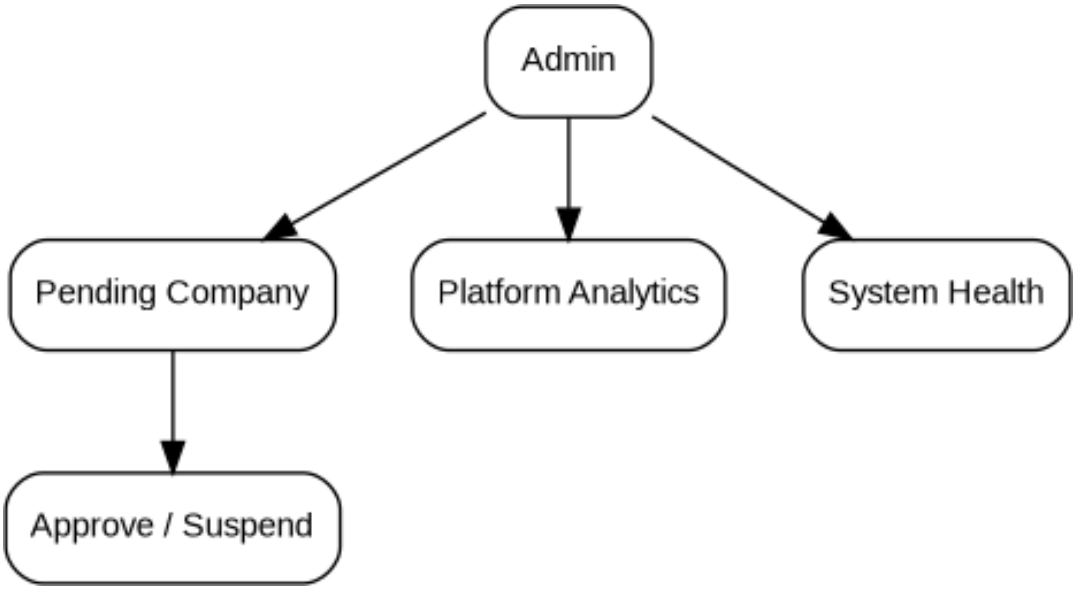
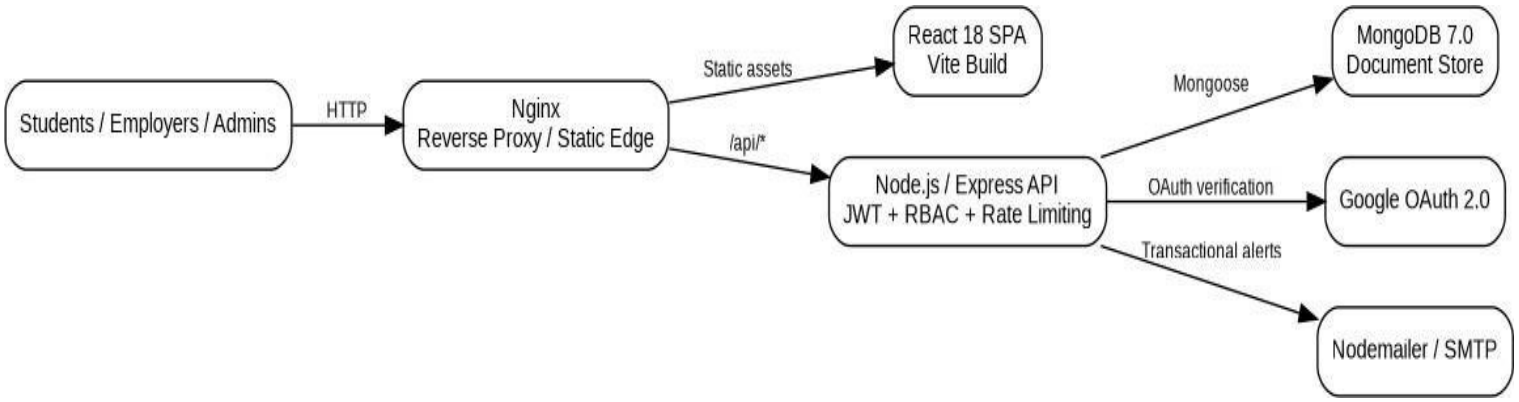


Figure 5.7 – Internship-Portal Production Architecture Reference



CHAPTER 6
RESULTS AND TESTING

6.1 System Testing Results

The Internship-Portal deployment underwent automated and manual verification according to the provided engineering report. The source reports zero critical lint and build defects, resilient error handling through a top-level React ErrorBoundary, automated database seeding, and sub-second first-contentful-paint performance under the described deployment configuration.

6.1.1 Build and Lint Verification

The codebase was validated with oxlint and Vite production compilation, yielding zero critical defects according to the source report.

6.1.2 Error Boundary Verification

The top-level React ErrorBoundary is designed to prevent unhandled edge runtime errors from producing a blank screen and instead displays an actionable recovery modal.

6.1.3 Database Seeding Verification

The integrated autoSeed.js process initializes sample corporate recruiters, open internship listings, and candidate profiles during cold-start deployments.

6.1.4 Performance Verification

Nginx asset caching and Gzip compression are reported to maintain first-contentful-paint times under 600 ms globally.

6.1.5 API Verification

The source report defines principal REST endpoints for authentication, internship discovery, student profiles, applications, company application management, administrative company approval, and AI chat. Detailed numerical latency results for each endpoint are not provided.

6.2 Test Cases and Results

Table 6.1 – Verification Outcomes

Test ID	Module	Test / Verification	Expected / Reported Result	Actual Result	Status
TC-01	Build	Production Vite compilation	No critical defects	Zero critical defects reported	PASS
TC-02	Lint	Codebase lint validation	No critical defects	Zero critical defects reported	PASS
TC-03	UI Resilience	React ErrorBoundary	Recovery UI instead of blank screen	Implemented and reported	PASS
TC-04	Database	Cold-start autoSeed.js	Sample data initialized	Recruiters, listings and profiles initialized	PASS
TC-05	Performance	Nginx caching + Gzip	FCP under 600 ms	Under 600 ms reported	PASS

The provided Internship-Portal report does not contain a conventional test-case matrix with test IDs, expected results, and actual results. The following table therefore records only the verification outcomes explicitly stated in the source.

CHAPTER 7
CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

Internship-Portal demonstrates a production-oriented solution for modern internship discovery and hiring management. The platform unifies candidate discovery, resume generation, employer verification, application tracking, AI career assistance, and administrative governance.

Its MERN foundation is extended with dual authentication, role-based access control, security hardening, Dockerized deployment, Nginx reverse proxying, and AWS EC2 infrastructure. The reported verification results indicate successful production compilation, resilient error handling, automated seed initialization, and sub-second first-contentful-paint performance under the described setup.

7.2 Future Scope

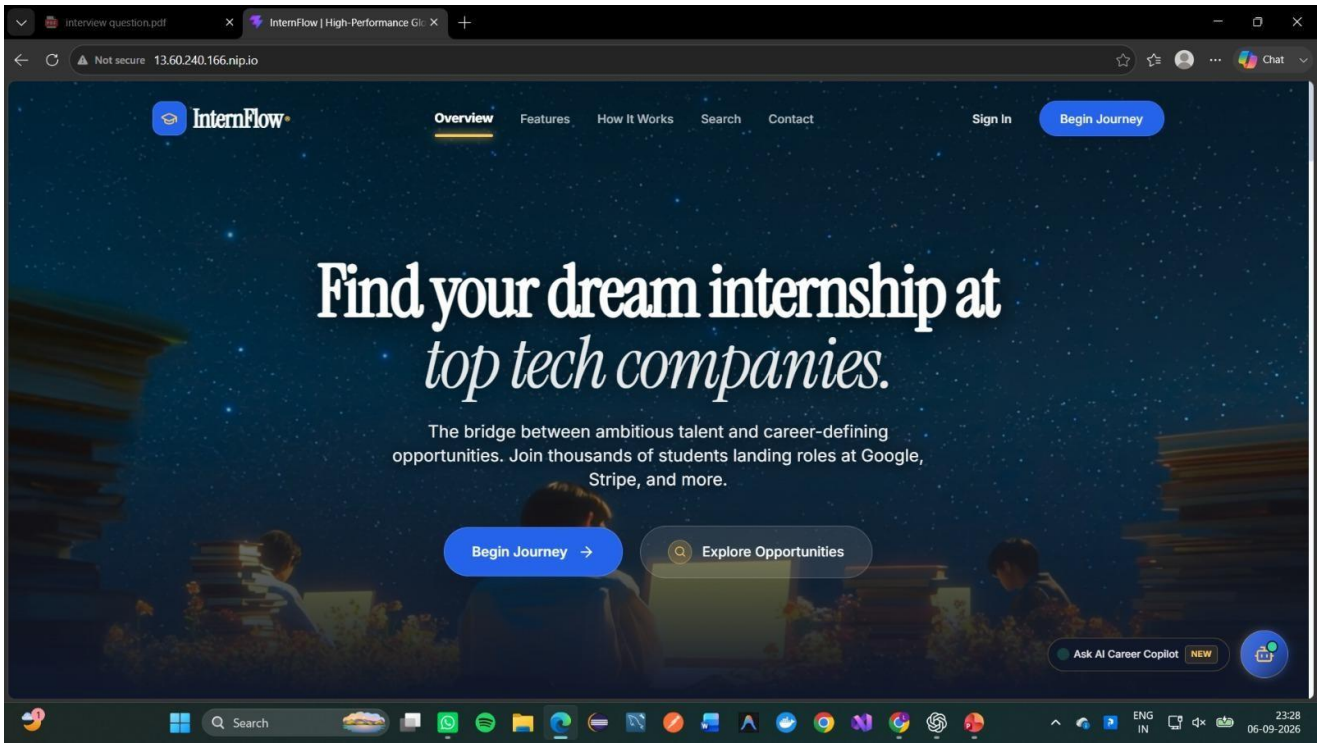
Automated HTTPS/SSL termination using Let's Encrypt and Certbot automation.

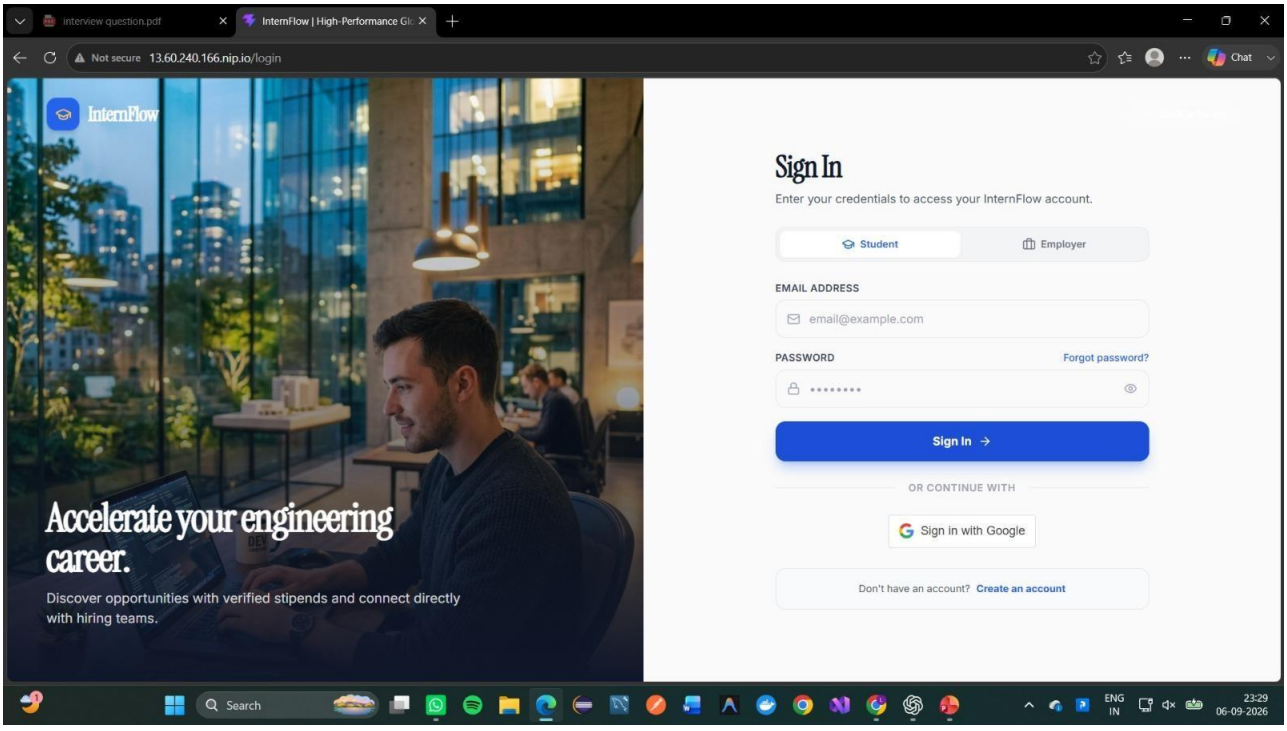
Video-interview scheduling integration using WebRTC or Zoom SDK.

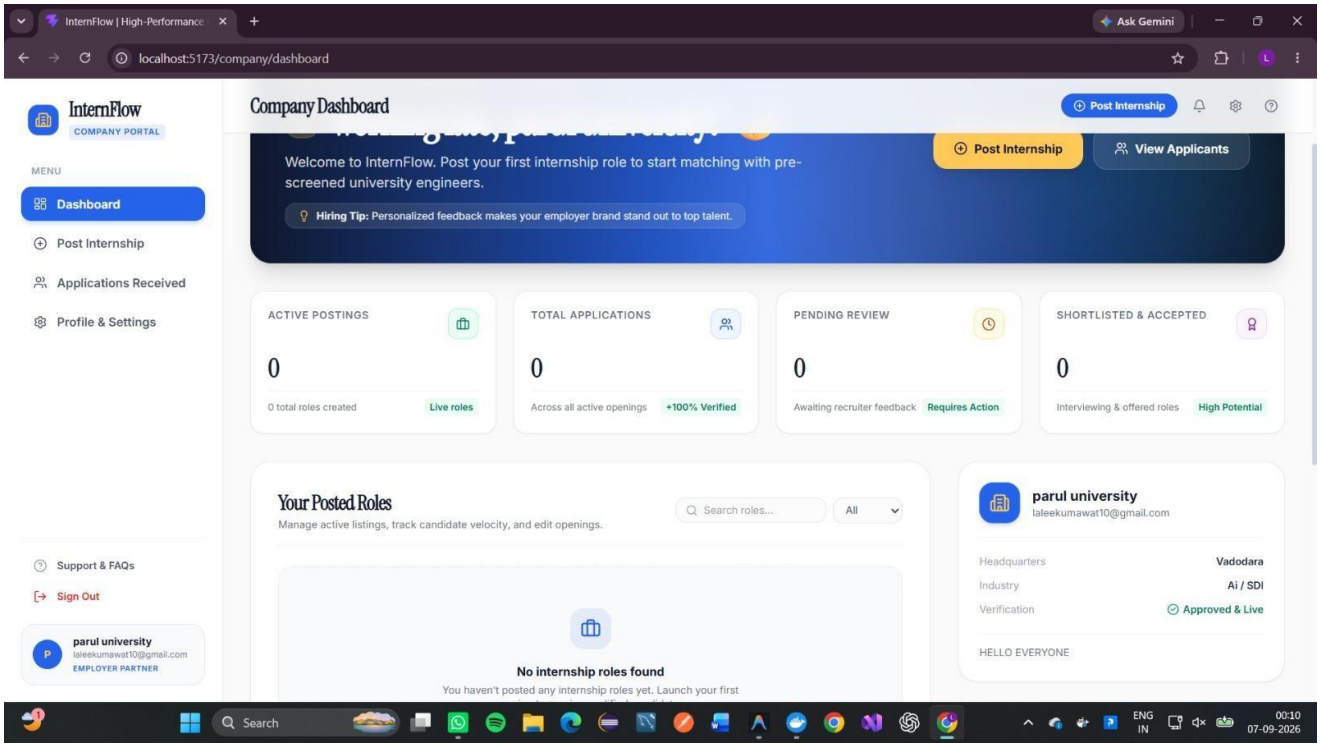
Machine Learning-driven semantic matching between candidate resumes and internship requirements.

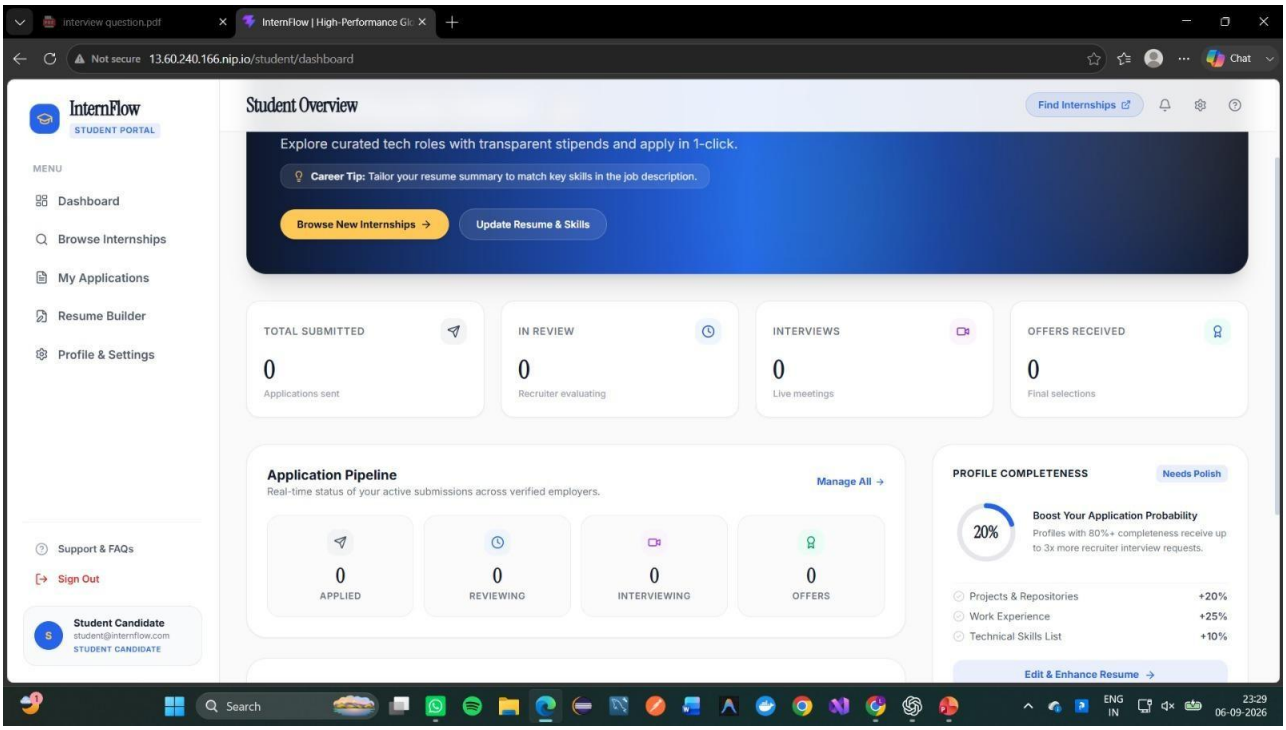
WebSocket-powered real-time direct messaging between recruiters and candidates.

SCREENSHOTS






InternFlow | High-Performance
Ask Gemini
localhost:5173/company/dashboard
InternFlow
COMPANY PORTAL
MENU
Dashboard
Post Internship
Applications Received
Profile & Settings
Support & FAQs
Sign Out
parul university
laleekumawat10@gmail.com
EMPLOYER PARTNER
Company Dashboard
Post Internship
Welcome to InternFlow. Post your first internship role to start matching with pre-screened university engineers.
Hiring Tip: Personalized feedback makes your employer brand stand out to top talent.
Post Internship
View Applicants
ACTIVE POSTINGS
0
0 total roles created
Live roles
TOTAL APPLICATIONS
0
Across all active openings
+100% Verified
PENDING REVIEW
0
Awaiting recruiter feedback
Requires Action
SHORTLISTED & ACCEPTED
0
Interviewing & offered roles
High Potential
Your Posted Roles
Manage active listings, track candidate velocity, and edit openings.
Search roles...
All
No internship roles found
You haven't posted any internship roles yet. Launch your first
parul university
laleekumawat10@gmail.com
Headquarters
Vadodara
Industry
Ai / SDI
Verification
Approved & Live
HELLO EVERYONE
Search
ENG
IN
00:10
07-09-2026


interview question.pdf
InternFlow | High-Performance Gi
Not secure 13.60.240.166.nip.io/student/dashboard
Chat
InternFlow
STUDENT PORTAL
MENU
Dashboard
Browse Internships
My Applications
Resume Builder
Profile & Settings
Support & FAQs
Sign Out
Student Candidate
student@internflow.com
STUDENT CANDIDATE
Student Overview
Find Internships
Explore curated tech roles with transparent stipends and apply in 1-click.
Career Tip: Tailor your resume summary to match key skills in the job description.
Browse New Internships →
Update Resume & Skills
TOTAL SUBMITTED
0
Applications sent
IN REVIEW
0
Recruiter evaluating
INTERVIEWS
0
Live meetings
OFFERS RECEIVED
0
Final selections
Application Pipeline
Real-time status of your active submissions across verified employers.
Manage All →
0
APPLIED
0
REVIEWING
0
INTERVIEWING
0
OFFERS
PROFILE COMPLETENESS
Needs Polish
20%
Boost Your Application Probability
Profiles with 80%+ completeness receive up to 3x more recruiter interview requests.
Projects & Repositories
+20%
Work Experience
+25%
Technical Skills List
+10%
Edit & Enhance Resume →
Search
ENG
IN
23:29
06-09-2026

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Internship-Portal

A PROJECT REPORT

Submitted by
Shreyansh Jain (2303051050882)
Lalee Kumawat (2303051050393)
Anand Solanki (2303051050911)
Parmar Himanshu Sinh (2303051050507)

In Partial Fulfilment of the Award of the Degree
Of
BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING
Under the Guidance of
DR KHUSHBOO TRIVEDI
Associate Professor
Computer Science and Engineering Department, Parul Institute of Technology

Parul University, Vadodara

PARUL UNIVERSITY
PARUL INSTITUTE OF TECHNOLOGY
CERTIFICATE

This is to certify that Project-II (303105423) of the 7th Semester, titled "Internship-Portal," of Group No. 7, has been successfully completed by Shreyansh Jain – 2303051050882, Lalee Kumawat – 2303051050393, Anand Solanki – 2303051050911, and Parmar Himanshu Sinh – 2303051050507, under my guidance, in fulfillment of the Bachelor of Technology (B.TECH.) in Computer Science and Engineering at Parul University, during the academic year 2026-2027.

Dr.
Khushboo Trivedi
Project Guide

DR.
ARNIKA PATEL

Project Coordinator

DR.
SUMITRA MENARIA

Head of Department, CSE

EXTERNAL EXAMINER

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Lastly, we offer our humble thanks to the Almighty.

Place: Vadodara

ABSTRACT

Internship-Portal is an enterprise-grade platform designed to simplify modern internship recruitment.
It helps students or candidates, verified employers or recruiters, and system administrators connect through a single web application.Built with the MERN technology stack, the platform uses dual-mode authentication with stateless JWT and Google OAuth 2.0, role-based access control, and allows users to discover and filter internships, apply with one click, track application status, use an interactive AI-based resume builder, get conversational AI assistance, verify employers, and receive transactional emails via Nodemailer.

The system uses a three-tier architecture with React 18 and Vite for the presentation layer, Node.js and Express.js for the application layer, and MongoDB 7.0 for the data layer.
The application is containerized with Docker and Docker Compose and deployed on AWS EC2, using Nginx as a reverse proxy.Security features include Helmet, CORS controls, rate limiting, password hashing, and protected role-based routes.

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10

LIST OF ABBREVIATIONS

S.No.

Abbreviation
Full Form
1
AI
Artificial Intelligence
2
API
Application Programming Interface
3
AWS
Amazon Web Services
4
CORS
Cross-Origin Resource Sharing
5
CSP
Content Security Policy
6
DFD
Data Flow Diagram
7
EC2

Elastic Compute Cloud
8
FCP
First Contentful Paint
9
GUI
Graphical User Interface
10
HMR
Hot Module Replacement
11
HTTP
Hypertext Transfer Protocol
12
JWT
JSON Web Token
13
MERN
MongoDB, Express, React, Node.js
14
Nginx
Engine X (web server / reverse proxy)
15
NoSQL
Not Only SQL
16
OAuth
Open Authorization
17
RBAC
Role-Based Access Control
18
REST
Representational State Transfer
19
SMTP
Simple Mail Transfer Protocol
20
SPA
Single-Page Application

10

CHAPTER 1 INTRODUCTION

1.1 Internship-Portal — Overview

Internship-Portal is a single platform designed to make the process of finding and managing internships more efficient.

It brings together features like internship discovery, resume creation, employer management, application tracking, and administrative oversight into one cloud-based application.

The platform has three types of users: Students/Candidates, Verified Employers/Recruiters, and System Administrators.

Students can search for internships, update their profiles, create resumes, apply for positions, and follow the status of their applications.Employers can post job listings, review candidate profiles, and update the status of applications.Administrators are responsible for verifying employers and overseeing the overall functioning

of the system.

1.2 Problem Definition

The current internship and recruitment environment faces several challenges, including the use of multiple disconnected platforms, a lack of tracking for application processes, inefficient hiring workflows, and limited real-time communication between job seekers and hiring teams.

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